

Systems Analysis and Design - RefHelp and Application Modelling

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1. Stakeholders and FURPS+

a) Determining Stakeholders for the RefHelp System

The RefHelp system seeks to simplify the allocation of referees to various matches in the **Rising Stars Football Association (RSFA)**.

Several relevant stakeholders to this system are:

1. RSFA Managing Committee

- They supervise the football association and make sure that the referee assignment system is rational, effective, and within the legal requirements of **IAFARA**.
- Their primary concern is ensuring the **integrity** of the **referee scheduling system**.

2. Volunteer Referees

- They are the people who officiate the games.
- They require that schedules are issued some reasonable time in the future, that they know which games they are scheduled for and that they can easily make a shift change if they need to be unavailable.

3. Football Clubs (Team Managers and Coaches)

- Clubs need to be certain that **referees are allocated in an impartial manner** so that there are no problems during the matches.
- They have to be able to retrieve information on the referees and make sure that the **officials fulfil the requirements set** (first aid, childcare clearance, etc.).

4. Parents and Guardians of Players

- As matches involve children, parents have an interest and concern in the **safety and impartiality** of the referees.
- They want to see trained and properly screened referees.

5. IAFARA (International Association Football Amateur Referees Association):

- An association of referees that deals with **amateur football associations** and integrates them into one body. This body guarantees that the referees have conducted their duties objectively and within the accepted **professional parameters**.
- They may request **reports or logs** to verify that referees are **being assigned indiscriminately**.

6. IT Team / System Administrators

- They will make sure that the **system runs without any flaws**, including operational and protection complications.
- They will deal with **bugs, upgrading software**, and anything related to **security**.

7. Government and Child Protection Authorities

- They ensure that referees are **permitted to interact with minors** by ensuring that all legal obligations are fulfilled.

- Could conduct audits to ensure that the referee author's documents are in order.

All of them help to achieve functional **smooth operations** within the **RefHelp system**. By meeting their needs, the system can become more **efficient, equitable, and compliant** in shifting the referee's roster.

b) Use Case Requirements of RefHelp

The functional requirements define what **RefHelp** must do to serve its users. Based on the provided case description, the system should have the following key features:

1. Referee Scheduling & Assignment

- Allocate referees and assistant referees **automatically** for any scheduled matches as they come.
- Allow referees to **indicate availability** in advance.

2. Volunteer Management

- Maintain records of each referee, including their **training status, certifications, and availability**.
- Notify referees about newly assigned matches and remind them of the games on the stipulated dates.

3. Match Information Tracking

- Maintain records of each referee and all the information related to the matches like **date, time, venue, and the relevant teams**.
- Allow referees to access and view scheduled match details from any web-enabled phone or computer.

4. Automated Alerts and Notifications

- Send SMS or email reminders **one week before, one day before, and on the morning of the match**.
- Allow referees to notify the system of their unavailability so that they can be replaced.

5. Substitution Management

- Allow referees who cannot officiate to **request for a substitute** to be allocated manually.
- Notify **available referees** and allow them to accept the substitution assignment automatically.

6. GPS Integration for Match Locations

- Provide a **map link** (like Google Maps) to enable referees to locate and navigate to their assigned match locations easily.

7. Secure Database & Access Control

- Store referee and match data securely with **role-based access** (Committee Members, Referees, IT Staff, etc).

- Ensure that all the personal data are password protected following proper security policies.

8. System Reports & Compliance Logs

- Produce audit reports for **IAFARA** as required including match **logs and referee assignments records**.
- Enable the managing committee to assess and interpret the referee's participation habits and trends.

By implementation of these functional specifications, will allow **RefHelp** to remove manual referee scheduling while promoting **equity, ease, and adherence** in the field.

c) Non-Functional Requirements Using FURPS+

Beyond its core functions, **RefHelp** must also meet several **non-functional requirements** that define its **quality, usability, and performance**. These requirements are categorized using the **FURPS+** model:

Functionality (F):

- The system must support **secure login** for different users (committee members, referees, IT staff).
- It should handle **multiple users accessing the system simultaneously** without issues.

Usability (U):

- The **web and mobile platforms** should be **user-friendly** and accessible to volunteers with no technical skills.
- Referees should also be able to see their schedules and update their availability with a **few clicks**.

Reliability (R):

- The system must have a **99.9% uptime** to ensure accessibility to referees at all times.
- It should also send **automated reminders**.

Performance (P):

- The **referee assignments** must be completed in less than 5 seconds.
- Notifications should also be **sent immediately** when the system is triggered.

Security (S):

- Match schedules and other personal details such as referees' information should be **encrypted**.
- Modification of referee assignments can only be done by **authorized users**.

+(Additional Considerations):

- **Design Constraints:** The system should be implemented using **web technologies compatible with** modern browsers and mobile devices.
- **Implementation Requirements:** Match data is to be stored in a **database like MySQL or PostgreSQL database**.
- **Interface Requirements:** The system is to be integrated with the **Google Maps API** to track locations.
- **Supportability Requirements:** The system should be able to accommodate a wider feature set in the future, for example, an AI system that assists in the balancing of active referee's workloads.

These requirements will help us ensure that **RefHelp** works, as well as is efficient, easy to use, secure, and dependable.

2. Use Case Modelling: User Goal Technique

a) Identifying Use Cases for the "Everything for Dogs" App

The **"Everything for Dogs"** application works towards helping dog owners search and book for different services like **training, grooming, walking, and boarding**. To probe deeper into the system's functional requisites, it is important to establish the principal features that users will be using in the application.

The User Goal Technique is the easiest method for pinpointing use cases because it concentrates on the goals users seek to achieve. We can identify specific goals for every user type by analyzing their expected interaction with the system.

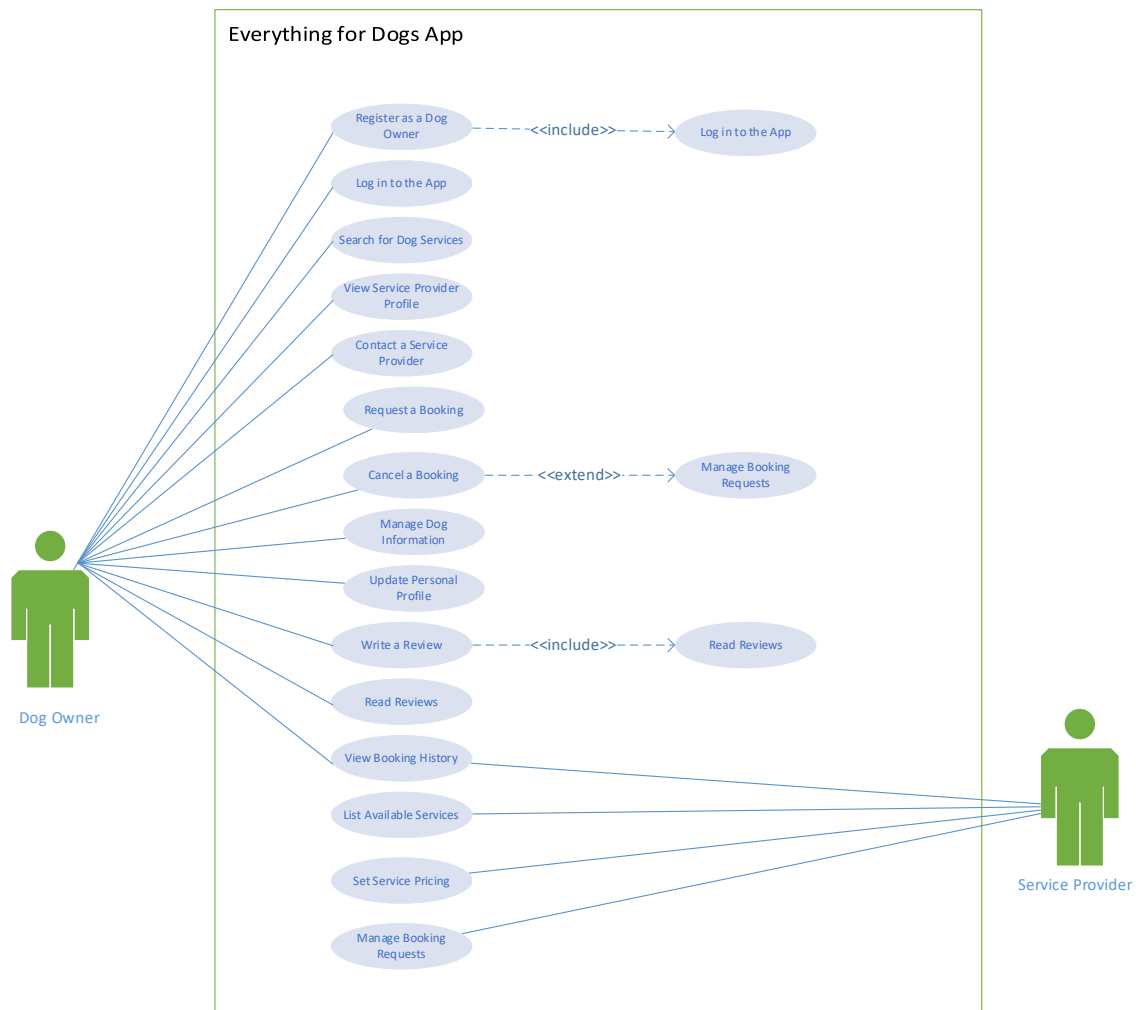
The **main users** of the app include:

1. **Dog Owners:** They utilize the system to search for service providers, make bookings, and tailor the services to their needs.
2. **Service Providers (Dog Trainers, Groomers, Walkers, and Boarders):** They provide the services and handle bookings.
3. **System Administrator:** They take care of user account management and system operations.

Use Case Table

Use Case Name	Description
Register as a Dog Owner	A new user (dog owner) signs up by providing personal details and pet information.
Log in to the App	The user enters credentials to access their account.
Search for Dog Services	The owner browses available services (boarding, grooming, walking, training) based on location and availability.
View Service Provider Profile	The owner views detailed profiles of service providers, including ratings, experience, and offered services.
Contact a Service Provider	The owner sends a message to a selected provider to ask about their services.
Request a Booking	The owner submits a booking request for a specific service at a chosen time.
Confirm a Booking	The service provider accepts or declines a booking request.
Cancel a Booking	The owner or service provider cancels a confirmed booking.
Manage Dog Information	Owners add or update details about their dog(s), such as breed, age, and special needs.
Update Personal Profile	The user modifies their account details, such as contact information or preferences.
Write a Review	Owners leave feedback for a service provider after completing a booking.
Read Reviews	Users browse ratings and reviews left by other customers.
List Available Services	Service providers create and update their listings (boarding, training, grooming, walking).
Set Service Pricing	Providers adjust their service prices and availability.
Manage Booking Requests	Providers accept, decline, or reschedule customer bookings.
View Booking History	Users check past bookings and service history.
Receive Notifications	The system sends alerts for booking confirmations, cancellations, and messages.
Reset Password	Users recover their account if they forget their password.
Deactivate Account	Users can delete or disable their account if they no longer wish to use the app.

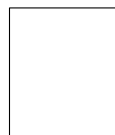
b) Use Case Diagram



LEGENDS



Actor



Boundary(System)



User cases



Association
Relationship



Include Relationship



Extend Relationship

3. Use Case Modelling – Event Decomposition Technique

Introduction

The **Yanchep Shielded Artificial Caving System (YSACS)** is designed to manage the operations of the **Yanchep Shielded Caving Centre (YSCC)**. It handles **bookings, staff management, cave maintenance, and safety protocols**.

To effectively model this system, we use the **Event Decomposition Technique**, which identifies **key events** that trigger system interactions. Events can be classified as **External, Temporal, or State** events:

External Events: A business process that can be initiated by an actor that is not controlled by the system (e.g., a customer or staff request).

Temporal Events: An event connected to a certain timestamp in the system's schedule (e.g., reminder notifications or tasks set up for a specific time).

State Events: An event is triggered when a system condition changes (e.g., the cave is 50 hours in use).

Event Decomposition Table

Event	Type of Event	Use Case	Brief Description	Actors (Only for external events)
Customer books a cave session	External	Book a Cave Session	A customer selects a date and time for a cave session enter participant details, and confirms the booking.	Customer
Customer cancels a booking	External	Cancel a Booking	A customer cancels their scheduled session. If cancelled within 4 weeks, no charge is applied.	Customer
The Customer receives booking reminders	Temporal	Send Booking Reminders	The system automatically sends SMS/email reminders 1 week and 1 day before the session.	System triggers, no actor
Customer signs statutory declaration	External	Submit Statutory Declaration	All customers must confirm they meet health and age requirements by signing a declaration before attending.	Customer
Customer books a 'wet' cave session	External	Validate Swimming Competency	The system ensures customers booking a 'wet' cave have a valid swimming certificate.	Customer
Staff checks cave safety before session	External	Perform Pre-Session Cave Check	A staff member checks for physical damage, cleanliness, lost belongings, and safety compliance before each session.	Staff
Equipment check before session	External	Perform Pre-Session Equipment Check	Staff inspect ropes, and breathing equipment, and ensure everything is safe for customers before the session starts.	Staff
Staff leads a cave session	External	Conduct Cave Session	A trained staff member guides customers through the cave while ensuring safety regulations are followed.	Staff
Cave session is completed	External	Log Session Completion	The system logs session time and updates cave usage hours.	Staff
Staff performs post-session cave check	External	Perform Post-Session Cave Check	After a session, the staff checks for damage, cleanliness, and hygiene issues before the next booking.	Staff

Equipment check after session	External	Perform Post-Session Equipment Check	Staff inspect ropes, and breathing equipment, and ensure everything is stored correctly after a session.	Staff
Customer requests session history	External	View Booking History	Customers can view their past cave bookings and session details.	Customer
Cave reaches 50 hours of use	State	Schedule Cave for Maintenance	The system marks the cave for servicing, removes it from the booking system, and schedules maintenance.	System-triggered event
Maintenance engineer inspects the cave	Temporal	Perform Scheduled Cave Inspection	An engineer inspects the cave daily before and after working hours.	Maintenance Engineer
Full cave service every 3 months or 50 hours	Temporal	Conduct Full Cave Maintenance	Every cave must undergo full servicing every 3 months or after 50 hours of use. The system logs maintenance completion.	System triggers scheduling
Drainage system inspection for 'wet' caves	State	Inspect Flood Drainage System	When a 'wet' cave is being serviced, the drainage system is also checked.	Happens during maintenance
System alerts staff about expiring safety training	Temporal	Send Safety Induction Reminder	The system notifies staff and the manager when a safety induction is expiring within a week.	System-triggered event
Staff undergoes safety induction training	External	Update Safety Induction Records	A staff member completes safety training, and the system updates their training record.	Staff
Generate cave usage report	External	Generate Cave Usage Report	The system generates a report showing cave bookings, popular session times, and trends.	Admin (Leslie)
Generate maintenance report	External	Generate Maintenance Report	The system provides an overview of upcoming maintenance schedules, completed servicing, and cave conditions.	Admin (Leslie)

Explanation of Event Types

The **event decomposition technique** ensures that every important **interaction or system process** is identified. Here's how the **different event types** play a role in the **YSACS system**:

1. External Events

- These involve **direct interactions** with customers, staff, or administrators.

- Examples: Booking a cave session, performing equipment checks, submitting safety declarations.

2. Temporal Events

- These happen at **fixed time intervals** and are handled automatically by the system.
- Examples: Sending booking reminders, notifying staff about training expiration, scheduling maintenance.

3. State Events

- These are triggered **when certain conditions are met**.
- Examples: When a cave reaches **50 hours of use**, it must be scheduled for maintenance.

4. Domain Modelling: UML Class Diagram

